

Maxwell Yang

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EDUCATION

Oregon State University

Corvallis, OR

Bachelor of Science, Computer Science

Expected June 2027 (**3.98 GPA**; Honors College Dean's List)

- Clubs: ACM (Association for Computing Machinery), OSU International Collegiate Programming Contest Team

EXPERIENCE AND RESEARCH

SWE Intern

May 2026 – July 2026

International Business Machines (IBM)

Baton Rouge, LA

Undergraduate Research Assistant

August 2025 - May 2026

Oregon State University

Corvallis, OR

- Designed a self-evolving prediction agent that improves accuracy by 4.4% through a dynamic **RAG** based system
- Implemented **vector embeddings** to retrieve past failure cases and synthesize guidelines for future predictions
- Designed a probabilistic weight update mechanism that automatically reinforces successful reasoning patterns and decays harmful biases based on Brier score feedback
- Integrated heterogeneous **LLMs** (OpenAI/Gemini) for specialized roles in reasoning and core prediction tasks

PROJECTS

Relational Database Management System | Python, MySQL

June 2025

- Designed a relational database schema normalized to **3NF**, enforcing referential integrity and foreign key constraints through DDL statements to eliminate redundancy
- Created single-column and composite indexes to reduce I/O overhead and improve read latency
- Managed database concurrency using explicit locks to eliminate dirty reads and maintain isolation, ensuring **ACID-compliant** transaction execution.

Transformer-based Autoregressive Language Model | Python, PyTorch, NLP

March 2026

- Implemented an autoregressive Transformer LM using **PyTorch**, trained to generate coherent children's stories
- Developed periodic positional encoding layers and integrated causal masking to ensure autoregressive properties during multi-head self-attention
- Engineered and compared decoding strategies, including Argmax, Temperature-scaled Sampling, and Nucleus (Top-p) Sampling, to optimize the balance between text coherence and creativity
- Evaluated model performance across varied prompts, analyzing the impact of temperature scaling and cumulative probability thresholds on distribution shift and output repetition

Workout & Nutrition Recommendation API | Python, FastAPI, PostgreSQL, HTML, JWT

July 2025

- Built a RESTful API that generates personalized workout programs and nutrition targets based on user goals, experience level, equipment availability, and dietary restrictions
- Added user authentication with **JWT**, enforcing per-user ownership and isolation of workout and nutrition plans
- Implemented full CRUD functionality, allowing users to create, view, update, and delete their own plans
- Persisted recommendation inputs and outputs using **PostgreSQL JSONB** for flexible and efficient data storage

Shell Implementation | Unix, C

April 2025

- Built a custom Unix-style command-line shell in **C**, supporting foreground/background process execution, I/O redirection, signal handling, variable expansion, and built-in commands
- Managed child process cleanup and zombie prevention, tracking background jobs and their exit status
- Implemented custom signal handling for SIGINT and SIGTSTP, allowing proper process termination.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, R, JavaScript, SQL, HTML/CSS, TypeScript

ML & Data Analytics: PyTorch, Autogen, Tableau, Power BI

Tools and Frameworks: FastAPI, Playwright, Vitest, Git, Linux/Unix, PostgreSQL

Techniques: LLMs, RAG, ML pipelines, RESTful APIs

ACTIVITIES

* ICPC Oregon Division 2 Winner, USA Computing Olympiad Gold Division, AIME Qualifier